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AI-Powered Chatbot Web Application

A Five-Day Full-Stack AI Development Project

Project Title: AI-Powered Chatbot Web Application

Live Website: https://first-website-chatbot.onrender.com/

GitHub Repository: https://github.com/ar-eej/First-website-Chatbot

LinkedIn: https://www.linkedin.com/in/ari18

Development Period: 18 February 2026 – 22 February 2026

Total Duration: 5 Working Days

Contents

Executive Summary

This assignment is about the design, development, testing and deployment of a full-fledged AI powered chatbot Web Application. The project has been completed in 5 working days and has proved the use of modern web technologies for the implementation of a useful real time communication system using artificial intelligence. The proposed application used different aspects such as HTML, CSS, JavaScript, Node.js, Express.js and OpenAI API. This application front-end has been developed with aim to access the responsive chat interface. However, the back-end manages the user's input that aligned with the OpenAI API, and ensures authentic responses. The last app was deployed on Render and live released under a web link. The goal of this project was to build a working chatbot and to learn the whole process of how to make a web application with AI technology. The success of final outcome have been seen when users ask questions and receive instant responses from developed chatbot. This project is also portfolio-worthy as it shows full stack development, working with APIs, securely using environment variables, and deploying to the cloud.

1 Introduction

There is no doubt regarding the fact that Artificial Intelligence is an essential part of the modern web application and major companies are using AI for developing error free applications. As the global shift can be seen where people are using chat bot for any query or use them as assistant for getting prompt responses. Here comes a chatbot in the picture, it allows to communicate normally with the system without clicking buttons and filling out forms. The goal of this project is to build a web-based chatbot that is able to converse with the user in real-time.The system is used to input a message by the user, call the Large Language Model with the message via the OpenAI API, return the generated response and present it in the chat window. This helps provide the feeling of a natural discussion with the AI. This project was chosen due to practical and logical reasons. A ChatBot melds together a variety of crucial aspects of computing today, such as frontend development, backend programming, API integration, cloud deployment, and artificial intelligence. The project also demonstrates that a valuable AI tool doesn't have to have a complicated user interface. With a robust back end and seamless API integration, a simple design can bring about a full and professional user experience.

2 Project Overview

AI-Powered Chatbot Web Application is the name of the project. It's a full stack web app allowing users to interact with an AI model through a chat web front-end. The main Chatbot function is simple. The user types a message in the input box, clicks the Send button and gets a message generated by the OpenAI language model. It's a real-time application, showing the user messages and AI responses in the clean chat window. The system has two major components as mentioned before the frontend responsible for visual interface and the interaction with the user. Although the backend receives messages, calls the OpenAI API and connect with frontend.

3 Project Aim

The main aim of this project was to design, develop, and deploy a web-based AI chatbot that can generate intelligent responses to user queries using a Large Language Model. This aim was chosen because the use of Chatbot technology applications are increasingly prevalent in Educational, Business, Customer Services, Personal Productivity and Digital platforms. Developing such a system provides hands-on experience in web development as well as integrating AI into web-based solutions.

4 *Project Objectives*

The main goals of the project:

1. To create a basic and simple chat interface.
2. To create a responsive design which will be usable on tablets and mobile/mobile and desktop.
3. Server side back end application with node.js and express.js.To connect the backend with the OpenAI API.
4. To secure API key with environment variables.

5 *Project Timeline*

Project Development Timeline											
Start Date:	2026-02-18										
End Date:	2026-02-22										
Duration:	3	Working Days									
							Day 1	**Day 2**	**Day 3**	**Day 4**	**Day 5**
WBS	**Task Description**	**Start Date**	**End Date**	**Duration**	**Status**	**Progress**	**18-Feb**	**19-Feb**	**20-Feb**	**21-Feb**	**22-Feb**
1.0	Planning & Design	2026-02-18	2026-02-18	1	Completed	100%					
2.0	Frontend Development	2026-02-19	2026-02-19	1	Completed	100%					
3.0	Backend Development	2026-02-20	2026-02-20	1	Completed	100%					
4.0	AI Integration & Testing	2026-02-21	2026-02-21	0	Completed	100%					
5.0	Deployment & Documentation	2026-02-22	2026-02-22	0	Completed	100%					
	Total Project Scope	**2026-02-18**	**2026-02-22**	**3**	**100% Done**	**100%**					

6 *System Design and Architecture*

System Design The chatbot is developed using a simple client-server paradigm. This implies that the user would be using the front end in the browser, with most of the processing being undertaken by the backend server. User message is passed to the backend by the frontend as a POST request. The message is sent to the backend, which then passes on to the OpenAI API. In that the OpenAI API returns a response, that can be seen on the backend as displayed messages. It can be explained as follows how the flow of the system is:

User → Frontend → Backend → OpenAI API → Backend → Frontend → User

7 Technologies Used

Multiple technologies were employed in this project, each for a specific purpose.

- The webpage was created using HTML, it generated the heading, chat container, input and send button.
- CSS was used to style the interface. It provided control of spacing, colors, message boxes, borders and layout design.
- The frontend was developed using HTML and JavaScript, where JavaScript was responsible for collecting user input, sending it to the backend for processing and displaying the result of the processing by the AI in the chat window.
- JavaScript on the server, using Node.js helped the project to get the back-end server up and running. It is a web framwork based with node.js (named Express.js). It simplified the process of making routes, managing requests and serving static files.
- Intelligence of the chatbot is provided by OpenAI API that offers the natural language responses to the user's input. The key was stored in an environment variable with dotenv, to hide the key in the code.
- All the version control, code repository and documentation were done via git and github.
- The application has been deployed on the internet using ‘Render' and hence it is available for everyone to use using a web browser. These technologies play a role in development of a completely integrated AI app with a full stack.

8 Step-by-Step Development Process

8.1 Planning and Design

8.2 Frontend Development

In this section the frontend development coding would be explained as developed with HTML, CSS and JS. The code for the front end of the project is listed below:

Smart logic in this front end ensures a smooth user experience as users don't have to refresh the page again and again that also maximize the time of user on the developed chatbot. Moreover this helps in messages getting adding instantly and the chat scrolls automatically to the latest reply.

8.3 Backend Development

This section is dedicated towards understanding the back-end of the application as it was developed with the help of Node.js and Express.js. The backend plays a crucial role in safeguarding the OpenAI API key. The codes on back end of the project are provided as follows:

9 AI Integration

One of the important parts of the project is the AI (Artificial Intelligence). The chatbot uses the API of the company 'OpenAI' to generate AI-based responses to the user's queries. The user's message is sent from user to backend and then to the GPT-3.5-turbo model. The model can read the input and generate the output in natural language form. The person will then see the response in the dialogue box. It's an integration to add functionality to the application other than the basic rule-based chatbot. The Rule Based Chatbot can answer only the questions or programmed rules. But this AI-based chatbot can understand and generate flexible responses for various questions.

10 Testing and Debugging

Testing was an integral part of the development process. Various types of input were performed to test the chatbot if it can respond correctly and smoothly. Functional checks were performed using simple questions. The quality of the AI's answer was tested with more complex questions. Empty messages were also tried to ensure that the system does not send needless requests. A check is added to the frontend code, so that blank messages would not be sent. Issues were tested related to deployment, too. This involved testing the frontend for successful loading, the backend route for proper response, and the OpenAI API key for its functionality on Render. Debugging enhanced the final reliability of application.

11 Deployment

The last deployed application was on Render. Render was chosen due to its simplicity to host a node.js application online. When deploying, the application files were linked from GitHub to Render. OpenAI API key was securely added to the environment variables in the Render dashboard. Once deployed, the test was carried out on the live website to ensure that the chatbot was functioning properly online. The live application can be found at: https://first-website-chatbot.onrender.com/ The deployment is a very significant milestone in this project as it brings the application from a local development project to become a real public application. That is, the user doesn't have to download or install anything. They just need to click on the link and open the link in the browser and use the chatbot.

12 Project Outcome

The end product of this project will be a fully functional chatbot web application powered by AI. The chatbot is accessible online and can reply to the messages via a neat browser interface. The following outcomes of the project were successfully achieved:

- Clean and modern chat interface was developed.
- Real time user interaction was integrated.
- Implemented backend using Node and Express.
- The integration with the OpenAI API was successfully implemented.
- Used environment variables to protect API key. The chatbot was subjected to various kinds of messages.
- The code was stored and documented on GITHUB. This is a meaningful project outcome as it shows technical competence and problem solving.

13 Challenges Faced

The project was successfully completed, but there were a few problems during development. One of the issues was getting the front and back end right. The front end needed to provide data in the appropriate JSON format, while the backend needed to accept the data in the proper format. Another challenge was handling the OpenAI API connection. Was missing correct API key, correct model name and correct request structure in the backend. There was also a need to pay attention to deployment, as environment variables needed to be set up individually on Render. A project can be successful locally, but not online due to incorrect deployment configuration settings. The other difficulty was to make it simple, yet useful. With only 5 days to build, the project had to be one of basic functionality without extraneous features. These challenges were overcome by testing and debugging and by an incremental development process.

14 Conclusion

Overall, this was a successful 5-day project and the team has created and deployed a working prototype of AI-powered chatbot web-application. It was built using HTML, CSS, JavaScript, Node.js, Express.js and the OpenAI API to create a realtime chatbot that users can interact with via a live website. This assignment demonstrates a clear and logical way in which one can develop a modern AI application. The chatbot was intended for planning. The user interface was developed using frontend development. Server logic was provided by backend development. We've added intelligence with the AI integration. Testing for better reliability. The application has been deployed and is now open to the public. The importance of the project is to demonstrate practical skills in full-stack web development and integration with artificial intelligence. It also shows the ability to have a successful technical project in a short period of time. Overall, this chatbot project is a good example of applied learning. It shows that with the right tools, planning and effort a full featured web app that makes use of artificial intelligence can be built in a very short time, professionally and effectively.

References and Project Links

Live Application:
https://first-website-chatbot.onrender.com/

GitHub Repository:
https://github.com/ar-eej/First-website-Chatbot

LinkedIn Profile:
https://www.linkedin.com/in/ari18